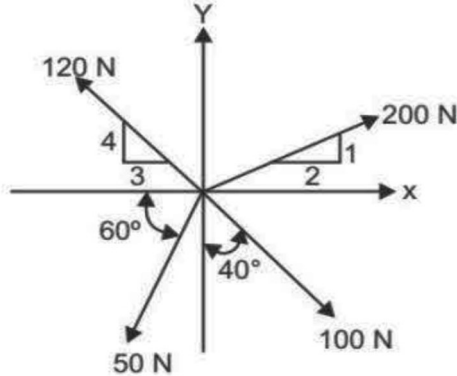
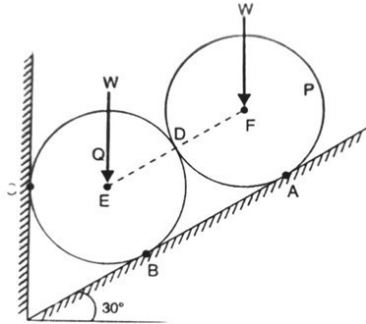
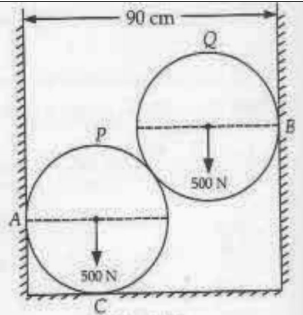
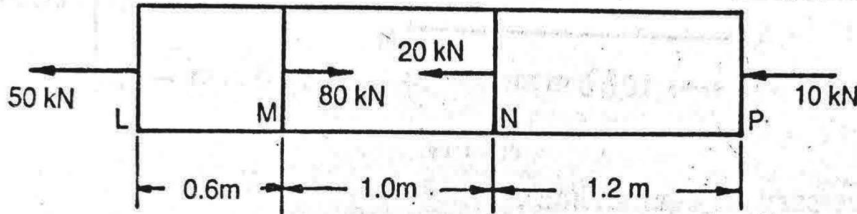
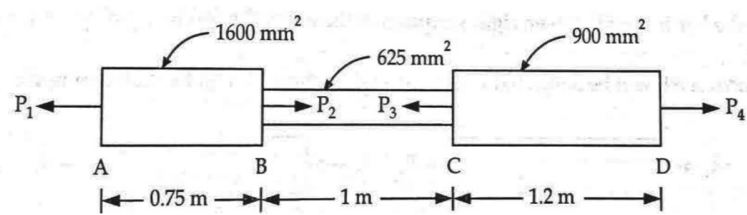
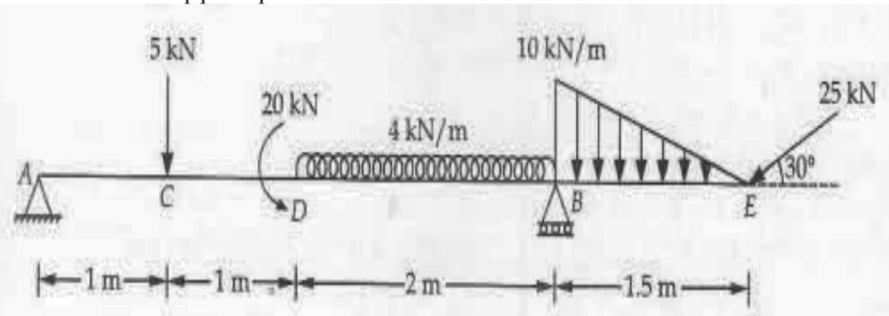
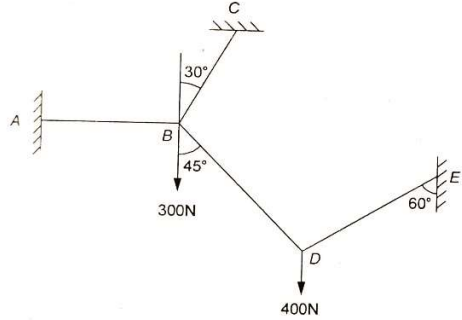
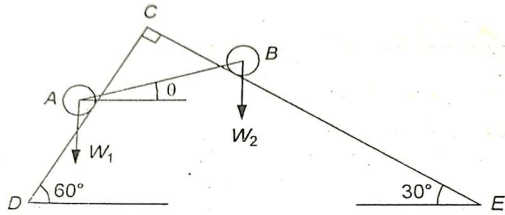


Question Bank Unit-I (Introduction to Mechanics and Strength of Materials)

Q.NO.	Short Answer type questions
1	Explain principle of transmissibility of force system.
2.	What are different types of force system?
3	What are various types of loads, beams and supports?
4	What is a statically determinate beam? Give some examples too.
5	Define Engineering stress and True stress.
6	How shearing stress is different from normal stress?
7	State and explain Hooke's law and poisson's ratio.
8	Discuss about superposition theorem.
9	What are elastic constants? Give their name.
10	Write formulae for relationship among elastic constants
Q.NO.	Short Answer type questions
1	A system of four forces acting on a body is as shown in figure. Determine the magnitude resultant and its inclination with positive X-axis. 
2	Two identical rollers P and Q, each of weight 100 N, are supported by an inclined plane and a vertical wall as shown in figure. Draw FBD of both sphere assuming all surfaces to be smooth. Also, find the reactions at points A, B, C and D. 
3	Two forces equal to P and 2 P respectively act on a particle. When the first force is increased by 120 N and the second force is doubled, the direction of the resultant remains the same. Determine the value of force P.
4	Two smooth spheres P and Q each of radius 25 cm and weighing 500 N, rest in a horizontal channel having vertical walls. if the distance between the walls is 90 cm, make calculations for the reactions on all the points of contact.

	
5	Draw and explain stress and strain curve for ductile material. Also, explain the difference between true stress and engineering stress.
6	Illustrate the various important points on stress strain diagram for mild steel. Also, explain the significance of different regions on the same.
7	A brass bar having cross sectional area of 1000 mm^2 is subjected to axial forces as shown in the figure. Find the total elongation of the bar and stress in individual components. Take Modulus of elasticity of Brass = 100 GPa 
8	A member ABCD is subjected to point loads P_1, P_2, P_3 and P_4 as shown in figure given below:  Calculate the force P_3 necessary for equilibrium if $P_1 = 120 \text{ kN}$, $P_2 = 220 \text{ kN}$ and $P_4 = 160 \text{ kN}$. Determine change in length of the member. Take modulus of elasticity 200 GN/m^2.
9	A beam has been loaded and supported as shown in figure given below. Determine the reactions at the support points A and B. 
10	Derive the relationship among E (Young's modulus), K (Bulk modulus) and μ (Poisson's ratio).
11	A square bar $50 \text{ mm} \times 50 \text{ mm}$ is subjected to a compressive load of 500 kN . The contraction over 200 mm length is 0.5 mm and increase in thickness is 0.04 mm . Calculate the value of the four elastic constants of the material.
12	At an axial load of 22 kN , a 45-mm -wide by 15-mm thick polyimide polymer bar elongates 3.0 mm while the bar width contracts 0.25 mm . The bar is 200 mm long. At

	the 22-kN load, the stress in the polymer bar is less than its proportional limit. Determine, 1. The modulus of elasticity. 2. Poisson's ratio. 3. The change in the bar thickness
13	A steel bar of square cross-section 35 mm x 35 mm, 500 mm long stretches 0.2 mm under a pull of 100 kN. The same bar in single shear test under a force of 122.5 kN shows a distortion of original right-angle corners by 0.00125 radian. Determine the values of the four elastic constants of the material.
14	Derive the relation between Young's modulus and Modulus of rigidity.
15	A vertical circular copper bar 20 mm diameter and 3 m long carries a tensile load of 200 kN. Calculate the elongation, decrease in diameter and the volumetric strain. Take $E = 100 \text{ GPa}$, Poisson's Ratio = 0.25
16	A metallic wire ($Y = 20 \times 10^{10} \text{ N/m}^2$ and $\mu = 0.26$) of length 3 m and diameter 0.1 cm is stretched by a load of 10 kg. Calculate the decrease in diameter of the wire.
17	A system of connected flexible cables is supporting to vertical forces of 300 N and 400 N as shown in the figure. Calculate the forces in various segments of the cable system. 
18	Two rollers of weight $W_1 = 100 \text{ N}$ and $W_2 = 200 \text{ N}$ are connected by a flexible string AB. The rollers rests on two mutually perpendicular planes CD and CE as shown in the figure. Find the tension in the string and the angle θ that it makes with the horizontal when the system is in equilibrium. Also find the reactions under the rollers. 
19	Two locomotives on opposite banks of a canal pull a vessel moving parallel to the banks by means of two horizontal ropes. The tensions in these ropes have been measured to be 20 kN and 24 kN while the angle between them is 60°. Find the resultant pull on the vessel and the angle between each of the ropes and the sides of the canal (AKTU 2024-25 even).
20	The magnitude of two forces is such that when acting at right angles produces a resultant force of 20 kN and when acting at 60° produce a resultant force of $\sqrt{28}$. Workout the magnitude of two forces.